

Nested Arnoldi Residual Closure for Directional Stein Tails

Repairing the One-Step Propagation Failure

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Abstract

RH-62 introduced an Arnoldi residual certificate for a directional Stein tail. Its projected component was sharp, but the remaining Arnoldi defect was propagated with an ordinary operator norm; on the RH-60 two-block model that one-step certificate was worse than the original norm envelope.

RH-63 expands the defect again. We prove a coherent nested identity in which all projected residual vectors are added before taking a norm, and only the terminal unexpanded remainder receives a positive norm bound. The resulting certificate is always an upper and is exact when the residual chain reaches Arnoldi breakdown. This is a finite-dimensional theorem, independent of normality.

At horizon $L = 32$, the RH-60 two-block model improves from a one-level gain of 66.41 to exactly 1 under a two-level schedule. A four-step nonnormal chain improves from 84.05 to 52.30, 53.48, and finally 1 at four nested levels. The RH-61 slow/fast surrogate is already nearly exact at one level, and recursive triangle splitting can make it slightly more conservative. The route remains viable, but uniform residual depth, terminal propagation, and block cross-column fusion are open. No physical-family, self-adjoint, arithmetic, or Hilbert–Polya conclusion is claimed.

Keywords: nested Arnoldi; Krylov residual; Stein tail; nonnormal operator; directional Hardy energy.

MSC 2020: 47A10; 47B65; 65F35; 93B07; 93D05.

1 Introduction

RH-61 identified a sharp difference between the actual directional tail and the largest-norm Stein envelope. RH-62 used the source Krylov space to capture part of that difference, but its residual term still used a worst-case norm. The two-block model made the limitation explicit: one Arnoldi vector gave a gain of about 66 although the full two-dimensional Krylov space was exact.

The defect is itself a vector with a structured orbit. Applying Arnoldi to that defect and retaining its projected orbit is the smallest coherent extension. The construction is related to standard nested Krylov ideas [1, 2], but the purpose here is a positive tail certificate rather than a linear-system solve.

Contributions and boundary

1. We prove the coherent nested Arnoldi identity.
2. We derive a positive terminal remainder bound.
3. We prove exactness when the residual chain terminates.

4. We audit slow/fast, two-block, and nonnormal-chain models and record the remaining depth and block-fusion barriers.

All model values are deterministic binary64 calculations. The RH-61-calibrated model matches a contraction gap only; it is not a stored folded-Gaussian production matrix.

2 Nested residual identity

Let A be a square matrix and $z \neq 0$. At level 0, an Arnoldi relation is

$$AV_0 = V_0H_0 + g_0e_{k_0}^*, \quad z = V_0\beta_0. \quad (1)$$

Apply the same construction to g_0 , obtaining

$$AV_1 = V_1H_1 + g_1e_{k_1}^*, \quad g_0 = V_1\beta_1, \quad (2)$$

and continue for a finite schedule. A level with $g_\ell = 0$ is a breakdown and terminates the chain.

Theorem 2.1 (Coherent nested identity). *For every horizon $L \geq 0$, recursively substituting the level relations gives*

$$A^Lz = \mathcal{A}_L + \mathcal{R}_L, \quad (3)$$

where $\mathcal{A}_L$ is the sum of every projected Arnoldi power generated at every level, with its exact complex coefficient, and $\mathcal{R}_L$ is a sum only of terminal unexpanded residual powers.

Proof. At one level, the identity is

$$A^Lz = V_0H_0^L\beta_0 + \sum_{j=0}^{L-1} A^{L-1-j}g_0e_{k_0}^*H_0^j\beta_0.$$

Replace each power of g_0 by its level-one identity. The projected level-one terms are added to the first term before taking a norm. Repeating the substitution at each finite level leaves only terms at the last unexpanded level. This is precisely (3). $\square$

Proposition 2.2 (Positive nested certificate). *Let $\rho \geq \|A\|$. If the terminal residual terms at level ℓ have coefficients $c_{\ell,j}$ and terminal sources g_ℓ , then*

$$\|A^Lz\| \leq \|\mathcal{A}_L\| + \sum_{\ell,j} |c_{\ell,j}| \rho^{u_{\ell,j}} \|g_\ell\|. \quad (4)$$

Here $u_{\ell,j}$ is the remaining power of the terminal residual. If the last residual vanishes, the second term is zero.

Proof. Use the exact coherent decomposition from Theorem 2.1, take one norm of the sum of all projected terms, and use $\|A^u g_\ell\| \leq \rho^u \|g_\ell\|$ only on the terminal terms. $\square$

Corollary 2.3 (Stein-tail form). *For $t_L = \sqrt{\kappa} \|S^Lz\|_{\mathfrak{S}_2}$, apply Proposition 2.2 columnwise and multiply by $\sqrt{\kappa}$. If every column terminates, the certificate is exact.*

Remark 2.4 (Remaining wall). *Nested substitution can increase the number of projected terms and still leaves ρ^u at the terminal level. Thus depth cannot be declared uniform without a residual-rank or residual-decay theorem. The construction is a route component, not Stage A1 closure.*

3 Finite-model audit

The pilot uses the same three models as RH-62. A schedule $(1, 1)$ means one Arnoldi vector at the source level and one at the residual level. The schedule $(1, 1, 1, 1)$ follows the four-step residual chain.

Table 1: Nested upper gains at horizon $L = 32$.

model	one level	two levels	full nested	full primary
RH-61 slow/fast surrogate	1.000000	1.012274	1.000000	1.000000
RH-60 two-block model	66.409450	1.000000	1.000000	1.000000
four-step nonnormal chain	84.053899	52.298902	1.000000	1.000000

The two-block row is the decisive repair: its first residual is an eigenvector, so the second level terminates and the projected terms reconstruct the exact vector coherently. The four-step row needs the full chain for exactness, while intermediate levels still reduce the error.

The slow/fast row illustrates a different issue. Its one-level certificate was already nearly exact because the initial vector has a very small slow component. A further triangle split can lose a small amount of coherence. Consequently, nesting is not monotonically better for every source; a stopping rule must compare the certified upper at each level.

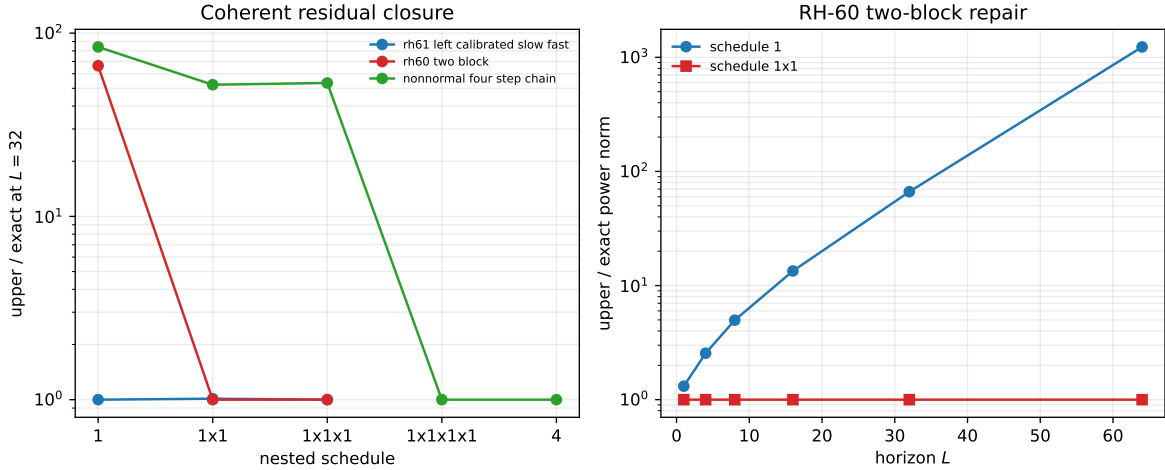


Figure 1: Nested residual audit. Left: the gain at $L = 32$ for successive residual schedules. Right: the two-block failure of one-level propagation and its exact repair by one nested level.

A 256-bit Arb calculation verifies the coherent two-level expansion for the RH-60 two-block model. The production physical operator is not interval validated.

4 Route consequence

RH-63 changes the route boundary identified in RH-62. The two-block counterexample is no longer a counterexample to residual recursion:

$$\text{source Krylov} \longrightarrow \text{residual Krylov} \longrightarrow \text{coherent projected sum} \longrightarrow \text{terminal positive tail.} \quad (5)$$

Three obligations remain before this can be transferred to the physical family:

1. bound the number of nontrivial residual levels uniformly or polylogarithmically;

2. replace the terminal ordinary norm by a weighted or directional residual certificate;
3. retain cross-column Gram information in a block version.

The next paper should address the second item on the smallest repaired model. If a weighted terminal certificate fails there, that failure will be a sharper obstruction than the original one-step norm gap.

5 No arithmetic or Hilbert–Polya conclusion

This paper constructs no self-adjoint operator, no $T \log T$ counting law, no von Mangoldt or prime-power trace formula, and no completed-zeta spectral identity. It makes no Hilbert–Polya or Riemann-hypothesis claim. The TPC branch is not used.

6 Reproducibility

The directory contains the recursive algebra, tests, model pilot, Arb audit, figures, hashes, and publication artifacts. The main commands are:

```
pytest -q -p no:cacheprovider
python experiments/run_nested_krylov_pilot.py
python experiments/run_arb_nested_audit.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The nested identity and positive terminal bound are analytic finite-matrix results. The tables are binary64 model evidence. Stage A1, Stage A4, uniform residual depth, block cross-column fusion, a self-adjoint Hilbert–Polya operator, a prime-power trace formula, and a zeta-zero identity remain open.

References

- [1] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, 4 edition, 2013.
- [2] Yousef Saad. *Iterative Methods for Sparse Linear Systems*. Society for Industrial and Applied Mathematics, Philadelphia, 2 edition, 2003.